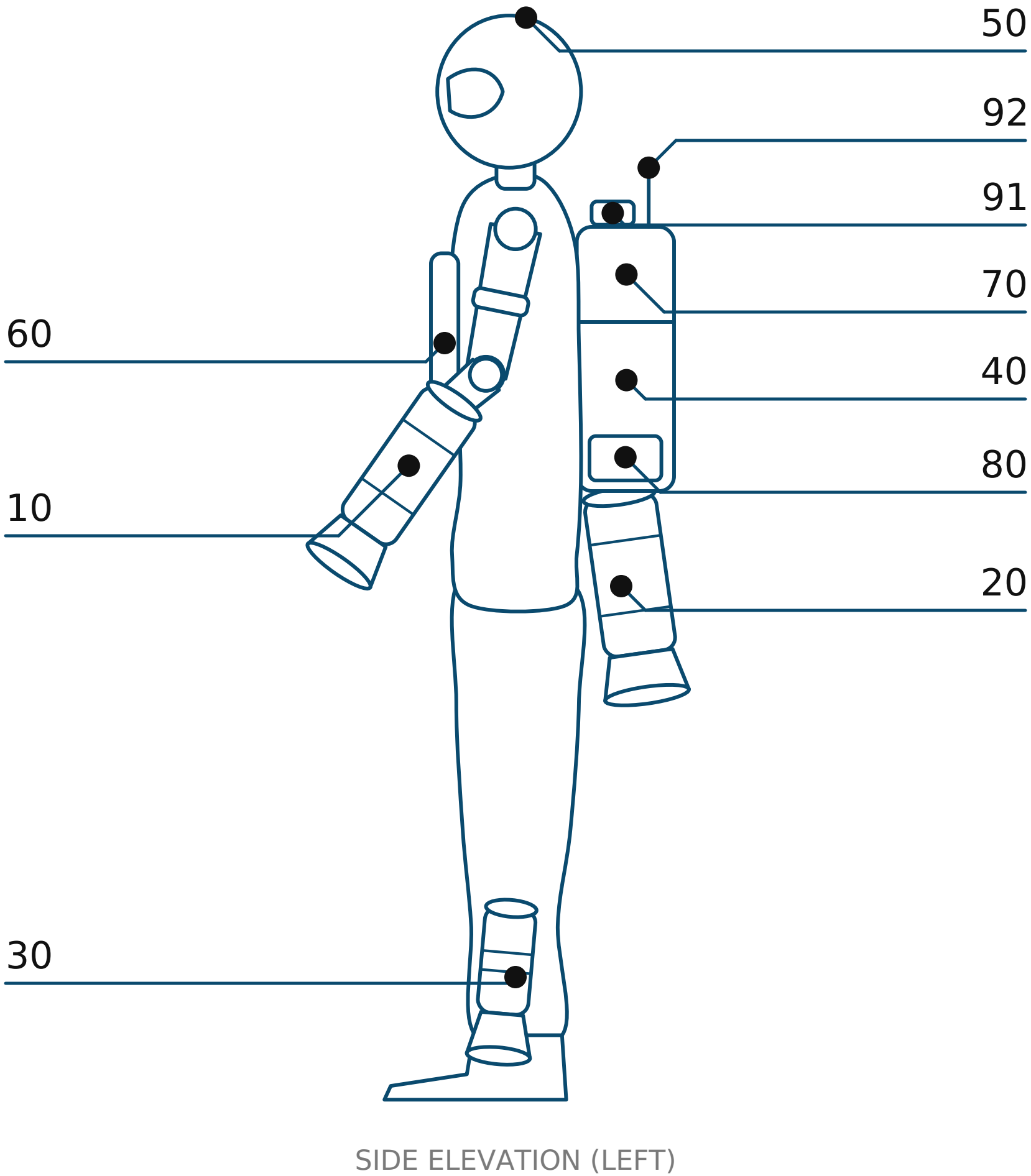
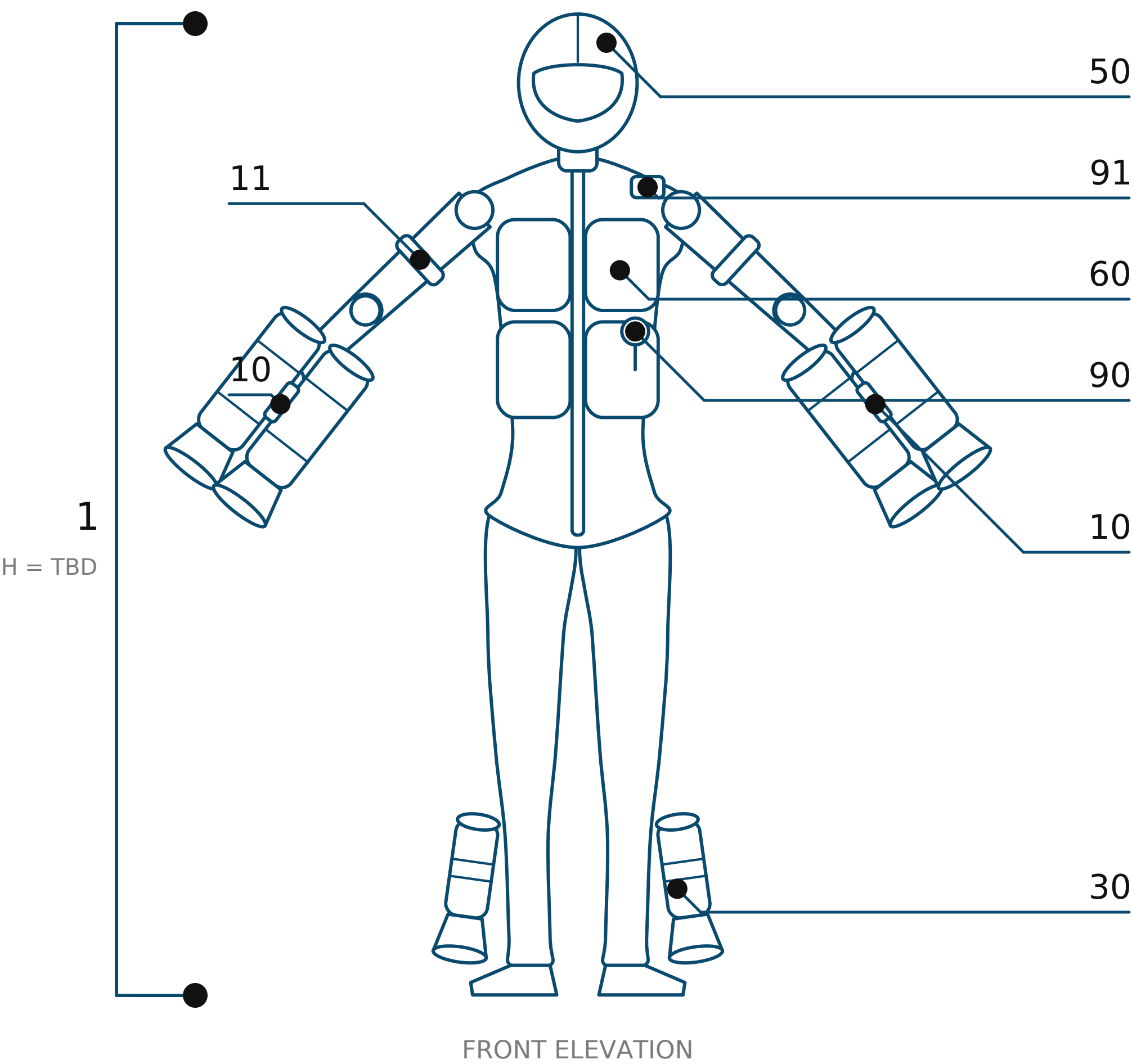


1 - General arrangement: front and side elevations

- NOTES
- 1. CONCEPT drawing. No dimension, mass, thrust or angle is shown: all TBD.
 - 2. Body proportions, vectors and on-body placement are ILLUSTRATIVE, not to scale.
 - 3. Module set (arm L/R, rear, ankle L/R): assumption A-003. Placement: A-008.
 - 4. Item numbers refer to the parts list (HFP-X scheme).



ITEM	DESCRIPTION	QTY	REFERENCE
1	Wearable flight system assembly	1	HFPX-SYS-ARC-001
10	Arm propulsion module	2 (L/R, A-003)	Vol 04
11	Arm mounting and articulation	2 (concept TBD)	Vol 03
20	Rear propulsion module	1 (A-003)	Vol 04
30	Ankle propulsion module	2 (L/R, A-003)	Vol 04
40	Fuel and energy storage (type TBD)	TBD (ISS-003)	Vol 05, Vol 15
50	Smart helmet with HUD	1	Vol 10

ITEM	DESCRIPTION	QTY	REFERENCE
60	Torso pods: computers, distribution	TBD (A-008)	Vol 08, Vol 15
70	Emergency recovery pack (type TBD)	1	Vol 13.11
80	Emergency battery (channel E)	1 (A-001)	Vol 15
90	Pilot controls and manual handles	1 set (ISS-015)	Vol 10, Vol 13
91	Sensor cluster (IMU, GNSS, etc.)	1 set	Vol 09
92	Comms / telemetry unit and antenna	1	Vol 11

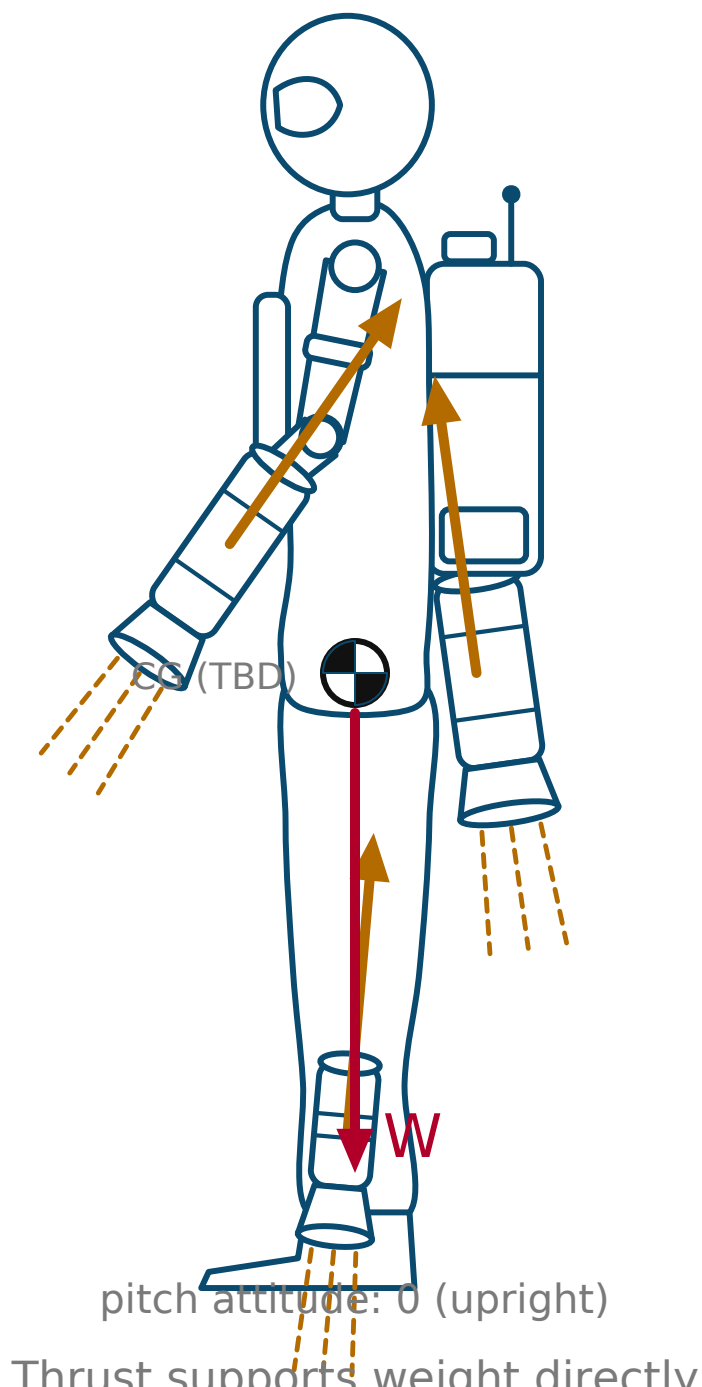
BOLTE HFP-X		Wearable flight system General arrangement	
HFPX-TDATA-DWG-001 Drawing ID		Sheet 1 of 13 Scale: NTS	
CONCEPT - not baselined (BL-0.0) Dimensions: TBD. Not for manufacture.		2026-09-29 Approver: TBD	

2 - Flight configurations: hover, transition, horizontal flight

- NOTES**
- 1. CONCEPT drawing. No dimension, mass, thrust or angle is shown: all TBD.
 - 2. Body proportions, vectors and on-body placement are ILLUSTRATIVE, not to scale.
 - 3. Module set (arm L/R, rear, ankle L/R): assumption A-003. Placement: A-008.
 - 4. Nozzle vectoring, control allocation and attitudes: TBD (ISS-006, ISS-009).
 - 5. Amber = module thrust (illustrative). Red = weight. Green = aerodynamic.

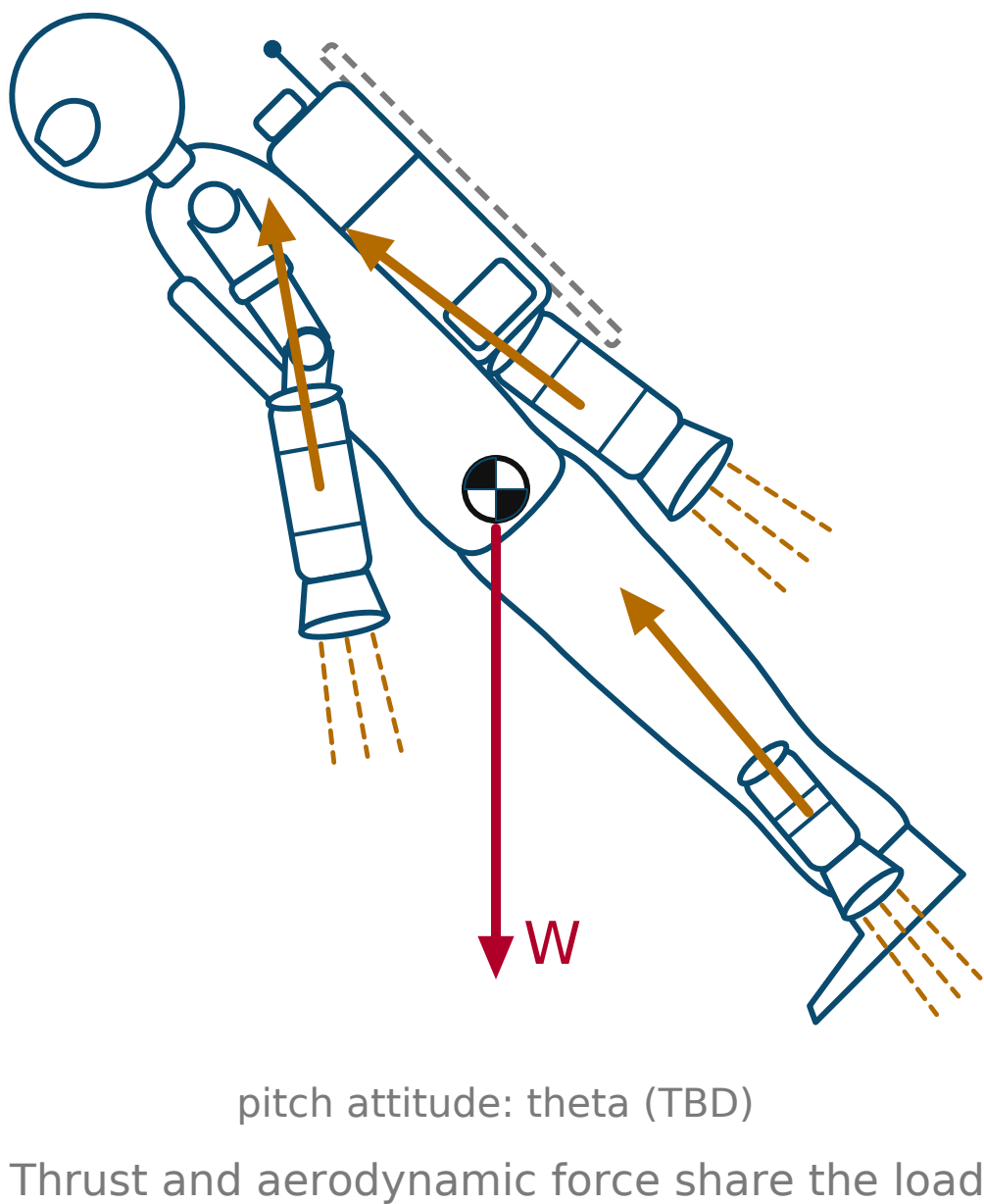
HOVER

(mode 4, HFPX-ARC-MOD-001)



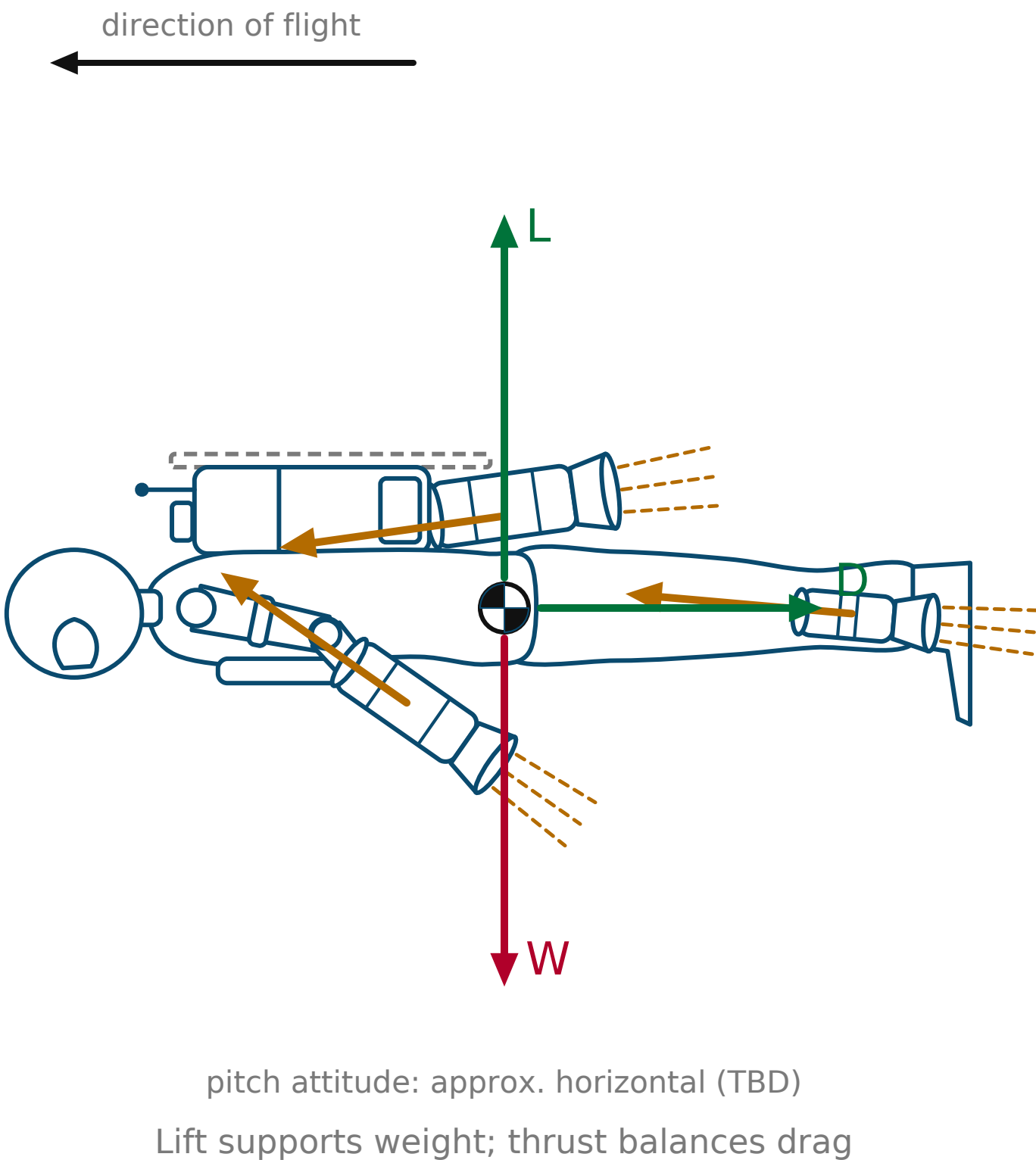
TRANSITION

(mode 6, HFPX-ARC-MOD-001)



HORIZONTAL FLIGHT

(mode 8, HFPX-ARC-MOD-001)



 Dashed outline = lifting surface (type TBD, ISS-004; whether it deploys: TBD).
Unmanned-first (DDR-001): the unmanned demonstrator carries no pilot; its mass properties must represent the piloted configuration (method TBC).

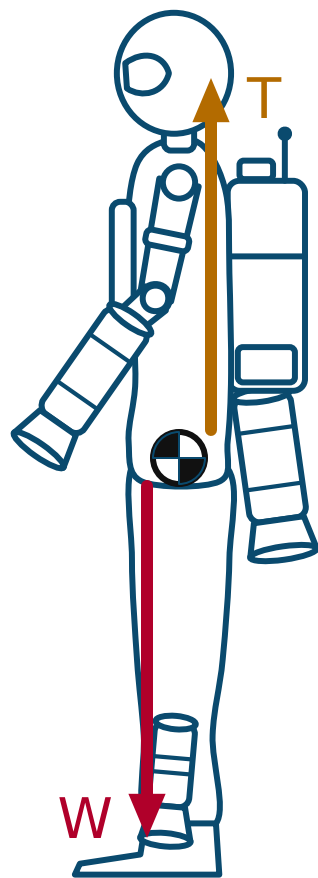
BOLTE HFP-X		Flight configurations Hover / transition / cruise	
HFPX-TDATA-DWG-002 Drawing ID		Sheet 2 of 13 Scale: NTS	
CONCEPT - not baselined (BL-0.0) Dimensions: TBD. Not for manufacture.		2026-09-29 Approver: TBD	

3 - Force and moment balance (free-body diagrams)

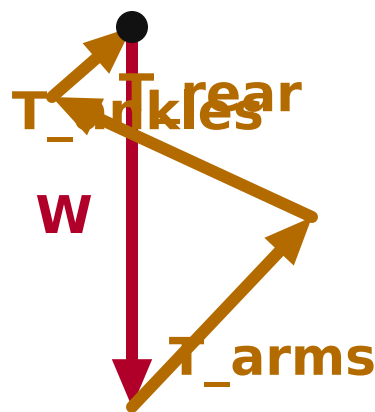
NOTES
1. SYMBOLIC ONLY. Vector lengths and angles are not to scale; no magnitude is claimed.
2. T: total thrust. W: weight. A: aerodynamic force (L lift, D drag).
3. Module thrusts $T_{i,j}$, i in {AL, AR, RR, NL, NR} (A-003), sum to T.

1 HOVER

quasi-static



closed force polygon (not to scale)

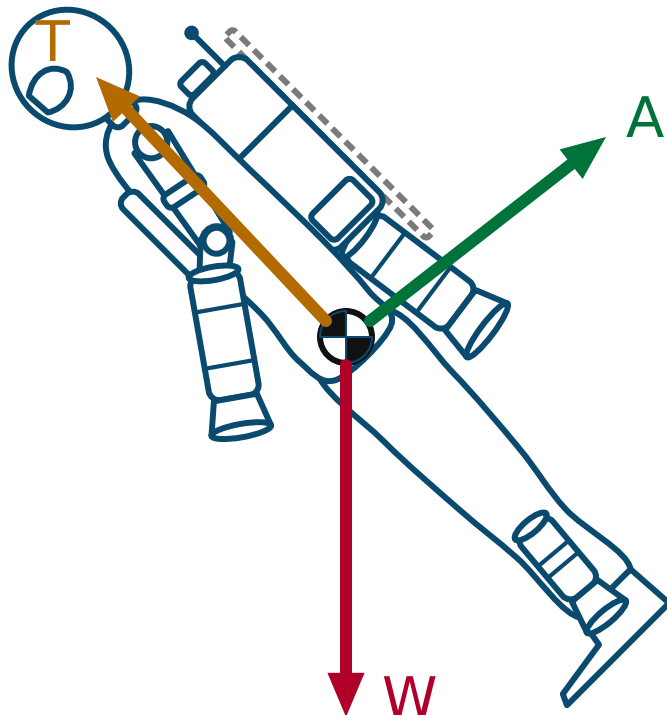


NEWTON'S 1ST LAW - steady flight

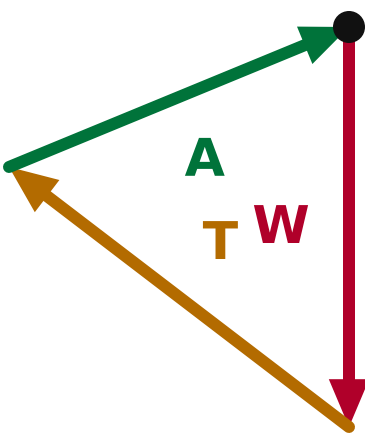
All forces balanced, net force zero:
 $\sum F = 0$ (closed force polygon)
Net moment about the CG zero:
 $\sum (r_i \times T_i) + M_{aero} = 0$
i.e. the resultant line of action passes through the CG

2 TRANSITION

quasi-static, illustrative



closed force polygon (not to scale)

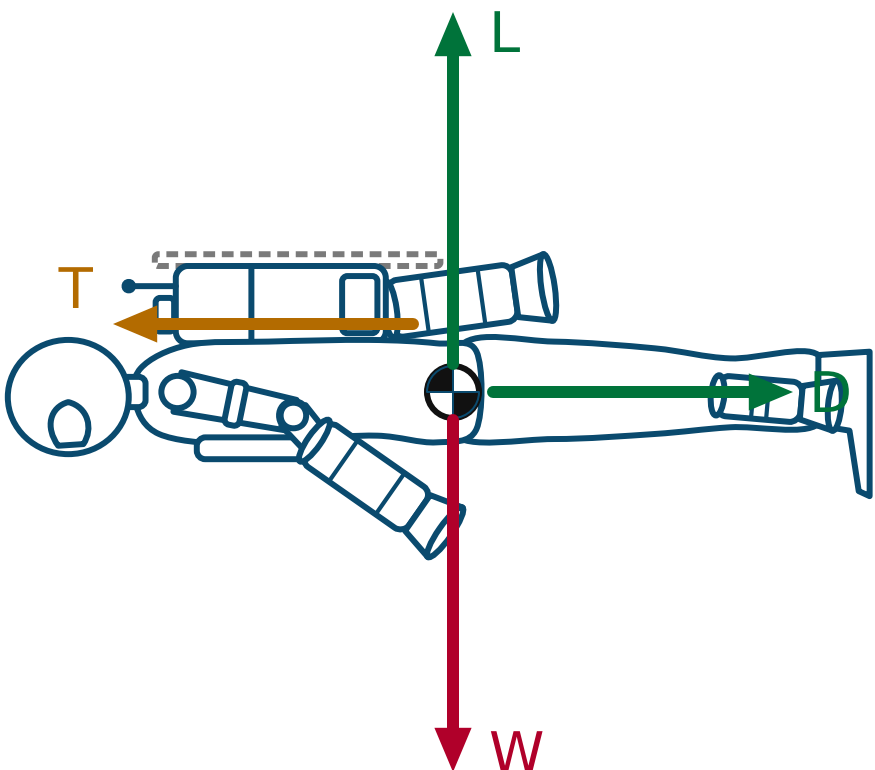


NEWTON'S 3RD LAW - one module

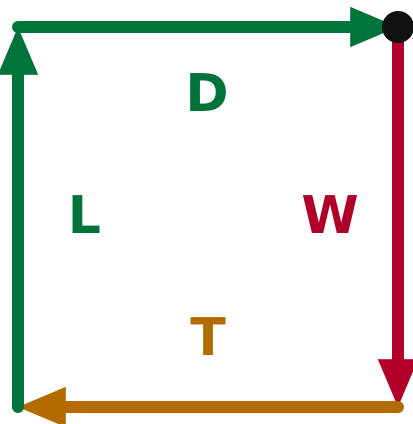
Force on exhaust by module = force on module by exhaust (equal and opposite).
Thrust, general form, one module:
 $T = m_{\dot{}} (v_e - v_0) + (p_e - p_0) A_e$
Magnitudes: TBD

3 HORIZONTAL FLIGHT

constant velocity



closed force polygon (not to scale)



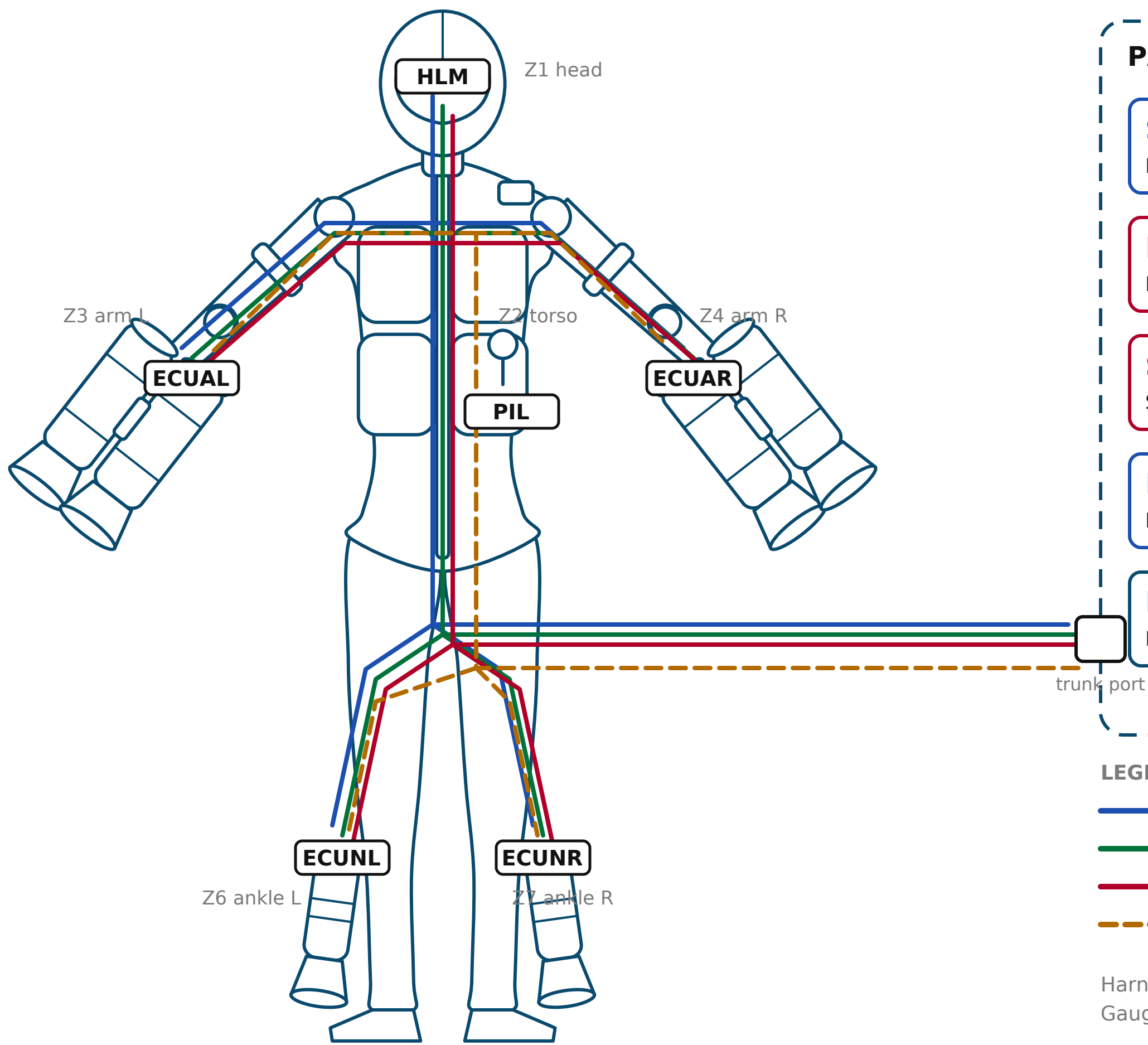
HOVER AND CONTROL CONDITIONS

Hover: $\sum T_{i,j} \geq W$ (thrust margin: TBD)
Trim and control need the moments too:
moment arms r_i of AL, AR, RR, NL, NR set control power
Mode-dependent allocation: TBD (ISS-006)

BOLTE HFP-X		Force and moment balance Free-body diagrams	
HFPX-TDATA-DWG-003 Drawing ID		Sheet 3 of 13 Scale: NTS	
CONCEPT - not baselined (BL-0.0) Dimensions: TBD. Not for manufacture.		2026-09-29 Approver: TBD	

4 - Installation schematic: zones, channel routing and fuel lines

- NOTES
- 1. CONCEPT drawing. No dimension, mass, thrust or angle is shown: all TBD.
 - 2. Body proportions, vectors and on-body placement are ILLUSTRATIVE, not to scale.
 - 3. Module set (arm L/R, rear, ankle L/R): assumption A-003. Placement: A-008.
 - 4. Routes are CONCEPTUAL. Physical separation of A/B/E is a design goal; routing TBD.



PACK / AVIONICS AND ENERGY BAY (Z8, Z9, Z5 - illustrative, A-008)

SRCA Primary source A	SRCB Primary source B	PDA Power distribution A	PDB Power distribution B
EMB Emergency battery	SCM Safety computer	FCA Flight computer A	FCB Flight computer B
SIM Safety-computer sensors	RCV Recovery controller	SNS Sensor cluster	COM Comms / telemetry
BB1 Data bus 1 backbone	BB2 Data bus 2 backbone	DLG Data logger	FMU Fuel management unit
ECURR Rear propulsion control	TANK Fuel storage (TBD)		

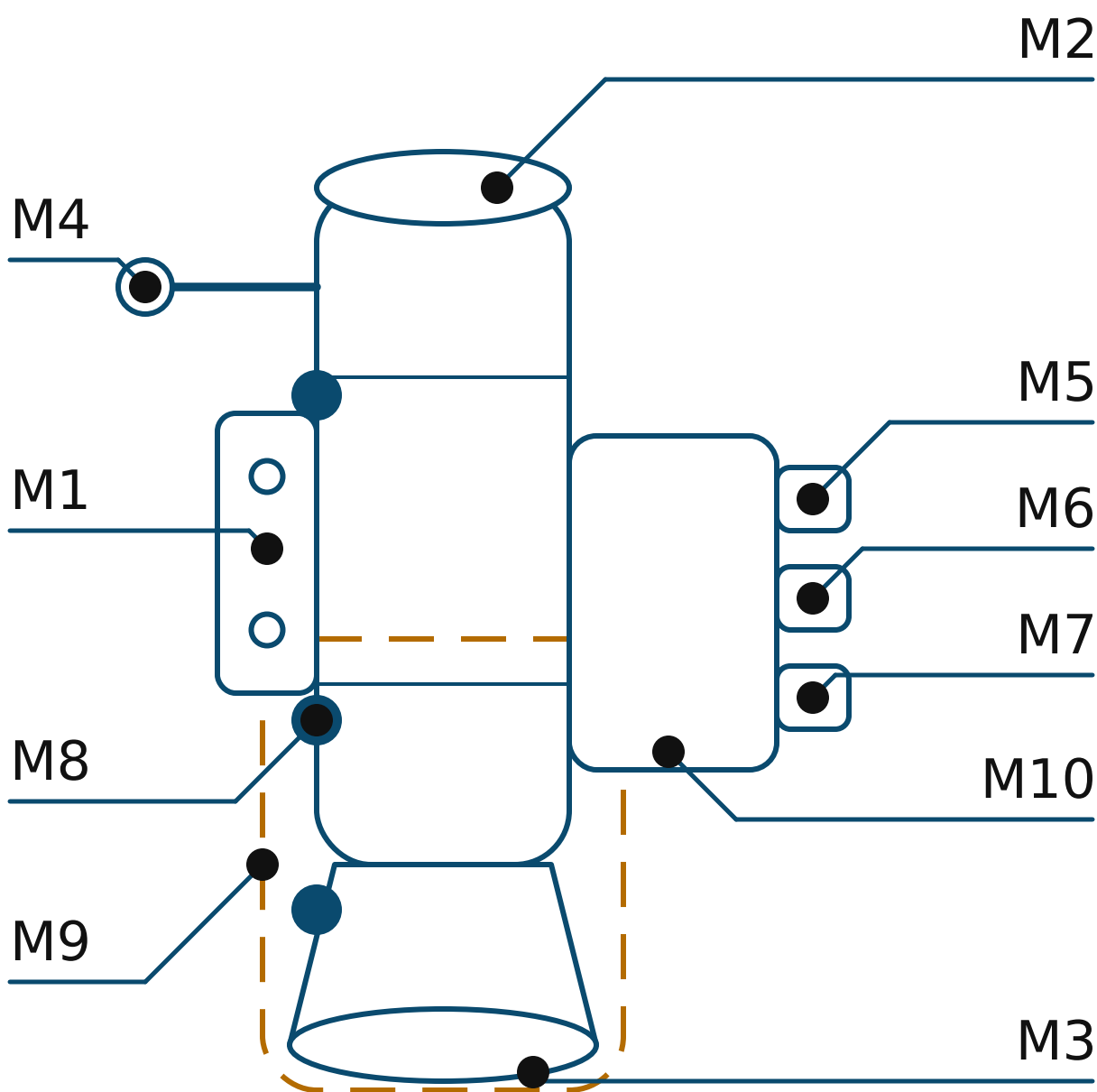
LEGEND

- Channel A: H-PWR-A-01 + H-DAT-A-01 (bus 1) [58 wires in wire_list.csv]
- Channel B: H-PWR-B-01 + H-DAT-B-01 (bus 2) [52 wires in wire_list.csv]
- Emergency E: H-PWR-E-01, H-DAT-E-01, H-DSC-E-01 [28 wires in wire_list.csv]
- Fuel line (architecture only: HFPX-FUEL-ARC-001)

Harness = one class, one channel (DR-W-02). Wire counts are read from wire_list.csv.
Gauge, insulation and zone temperatures: TBD (ISS-011, ISS-012).

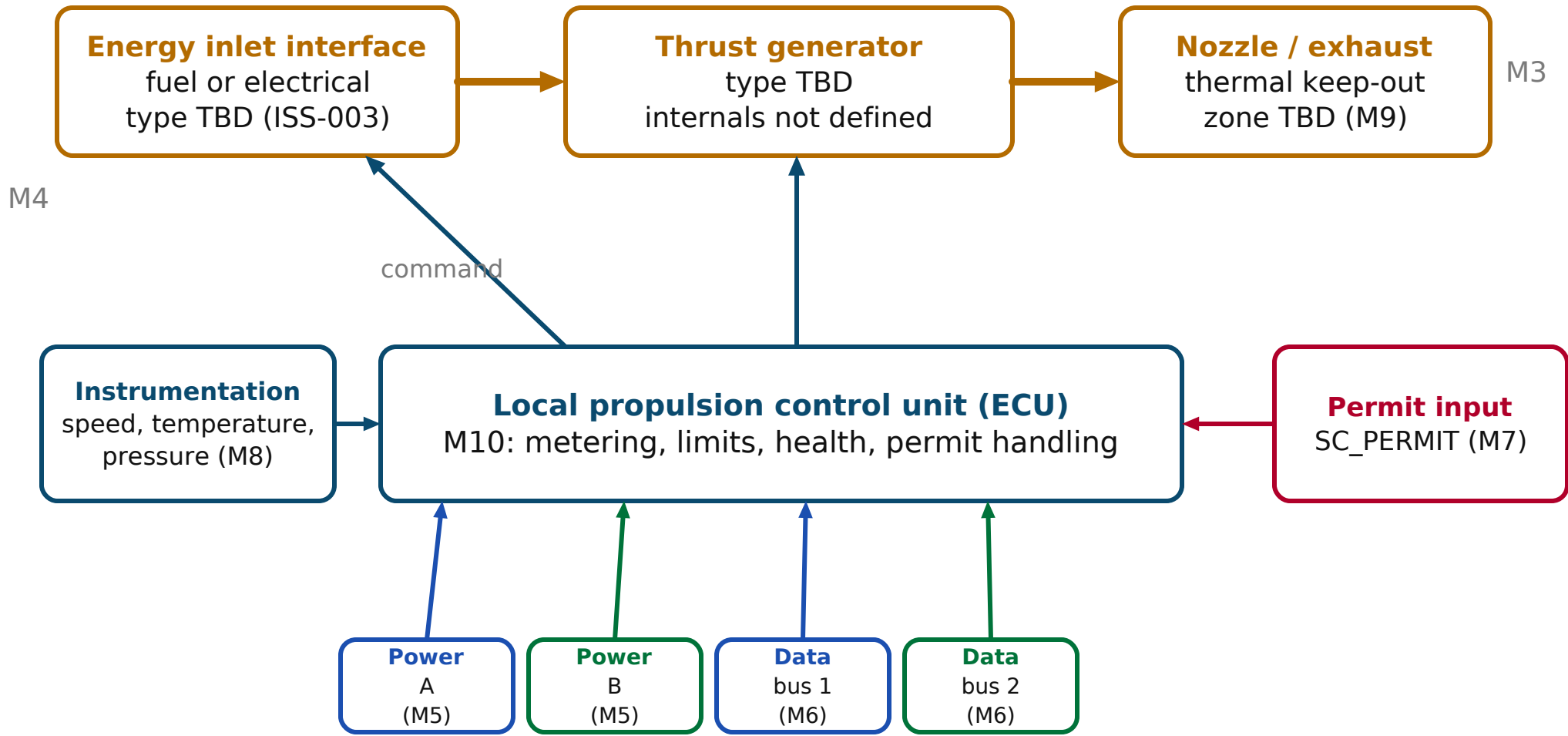
BOLTE HFP-X	Installation schematic Zones and channel routing	
HFPX-TDATA-DWG-004 Drawing ID	Sheet 4 of 13 Scale: NTS	
CONCEPT - not baselined (BL-0.0) Dimensions: TBD. Not for manufacture.	2026-09-29 Approver: TBD	

5 - Propulsion module: interfaces and functional schematic



MODULE ENVELOPE (side elevation, interface side)
One common interface definition for all five modules is desirable (TBD).

FUNCTIONAL SCHEMATIC (black box)



Mount (M1) carries thrust into the structure: see sheet 007. Thermal management: TBD (Vol 14).

MODULE VARIANTS AND CONNECTORS (allocated pins in brackets; from connector_list.csv)

Module	Zone	ECU	Power A	Power B	Data 1	Data 2	Permit
Arm left (AL)	Z3	ECUAL	J-ECUAL-PA (2)	J-ECUAL-PB (2)	J-ECUAL-D1 (2)	J-ECUAL-D2 (2)	J-ECUAL-DS (2)
Arm right (AR)	Z4	ECUAR	J-ECUAR-PA (2)	J-ECUAR-PB (2)	J-ECUAR-D1 (2)	J-ECUAR-D2 (2)	J-ECUAR-DS (2)
Rear (RR)	Z5	ECURR	J-ECURR-PA (2)	J-ECURR-PB (2)	J-ECURR-D1 (2)	J-ECURR-D2 (2)	J-ECURR-DS (2)
Ankle left (NL)	Z6	ECUNL	J-ECUNL-PA (2)	J-ECUNL-PB (2)	J-ECUNL-D1 (2)	J-ECUNL-D2 (2)	J-ECUNL-DS (2)
Ankle right (NR)	Z7	ECUNR	J-ECUNR-PA (2)	J-ECUNR-PB (2)	J-ECUNR-D1 (2)	J-ECUNR-D2 (2)	J-ECUNR-DS (2)

- NOTES
- 1. CONCEPT drawing. No dimension, mass, thrust or angle is shown: all TBD.
 - 2. Body proportions, vectors and on-body placement are ILLUSTRATIVE, not to scale.
 - 3. Module set (arm L/R, rear, ankle L/R): assumption A-003. Placement: A-008.
 - 4. INTERNALS of the thrust generator are NOT defined (type undecided, ISS-003).
 - 5. Interface positions are illustrative; connector data is read from connector_list.csv.

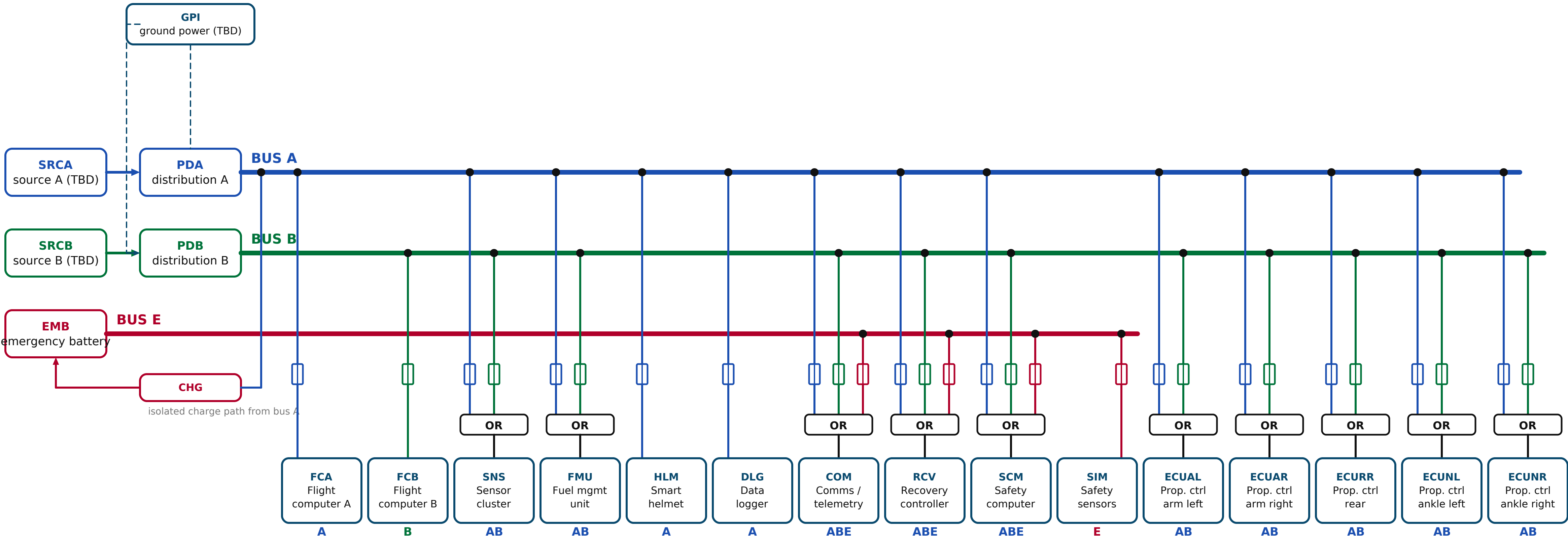
BOLTE HFP-X	Propulsion module Interfaces, functional	
HFPX-TDATA-DWG-005 Drawing ID	Sheet 5 of 13 Scale: NTS	
CONCEPT - not baselined (BL-0.0) Dimensions: TBD. Not for manufacture.	2026-09-29 Approver: TBD	

6 - Electrical single-line diagram

- NOTES
1. CONCEPT. All ratings TBD. F = feeder protection. OR = ideal-diode ORing (circuit c1).

2. Feeders are generated from hfp_x_wiring.FEEDS. Load classes are placeholders (A-006).

3. Crossing lines without a dot are NOT connected.

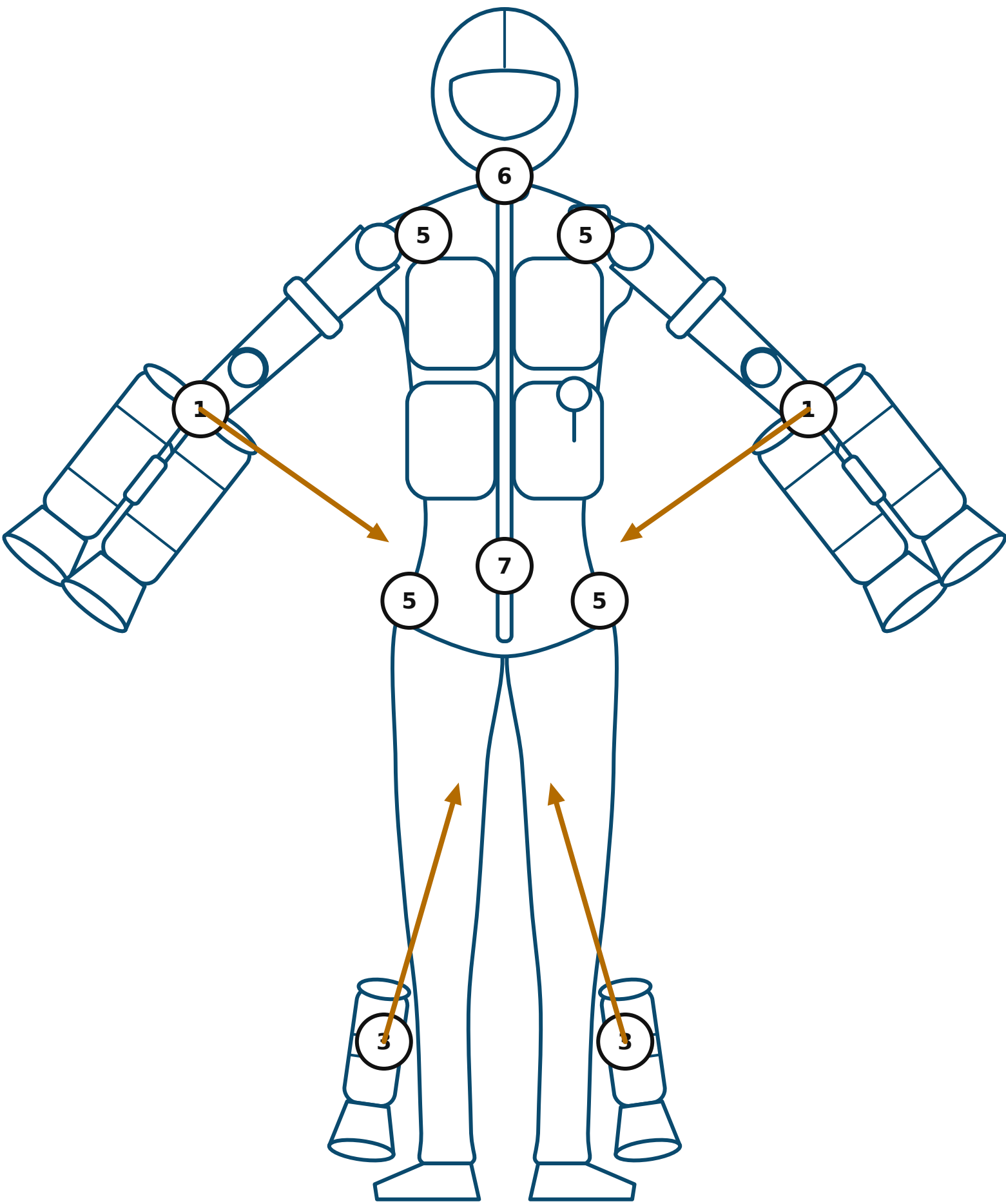


Feed channels under each load (A / B / E). A load with two or more feeds combines them locally through ideal-diode ORing.
Emergency channel E feeds only SCM, RCV, COM and SIM (A-001). The emergency battery is charged from bus A through an isolated path.
Bus tie between A and B: not drawn (open trade, HFPX-ELE-ARC-001).

 F feeder protection, rating TBD

BOLTE HFP-X		Electrical single-line Sources, buses, loads	
HFPX-TDATA-DWG-006 Drawing ID		Sheet 6 of 13 Scale: NTS	
CONCEPT - not baselined (BL-0.0) Dimensions: TBD. Not for manufacture.		2026-09-29 Approver: TBD	

7 - Structural load paths and mounting interfaces



Amber: thrust-reaction path from each module to the pelvis: route TBD (ISS-017)
LP2 (rear module mount) and LP4 (pack mount) are on the back, not visible here.

- NOTES
1. CONCEPT drawing. No dimension, mass, thrust or angle is shown: all TBD.

2. Body proportions, vectors and on-body placement are ILLUSTRATIVE, not to scale.

3. Module set (arm L/R, rear, ankle L/R): assumption A-003. Placement: A-008.

4. No load, limit or margin is stated. Load cases and factors: TBD (Vol 03).

5. HOW thrust is reacted into the pilot's body is unanalysed (ISS-017).

MOUNTING AND HARD POINTS

- 1

LP1 Arm module mount (L/R)
- 2

LP2 Rear module mount (back)
- 3

LP3 Ankle module mount (L/R)
- 4

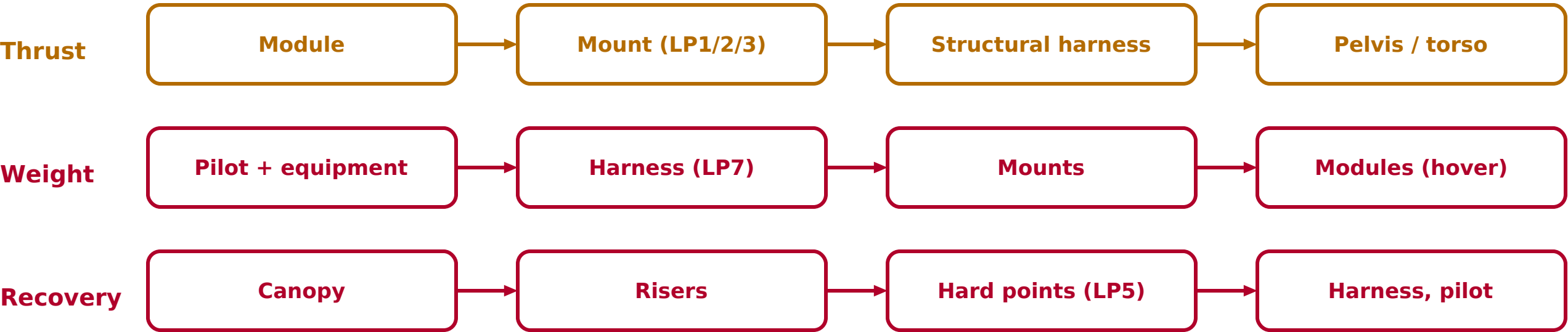
LP4 Pack mount (back)
- 5

LP5 Recovery riser attachments (shoulder, hip)
- 6

LP6 Helmet-to-harness interface (neck)
- 7

LP7 Pilot restraint / harness attachment

LOAD-PATH CHAINS (concept)



The recovery load path must not depend on the propulsion modules or their mounts (candidate design intent, to analyse).

LOAD CASES (none defined yet)

Load case	Load	Source
Ground handling	TBD	TBD
Hover	TBD	TBD
Manoeuvre and gust	TBD	TBD
Transition	TBD	TBD
Landing impact	TBD	TBD
Recovery deployment	TBD	TBD
Emergency (loss of module)	TBD	TBD

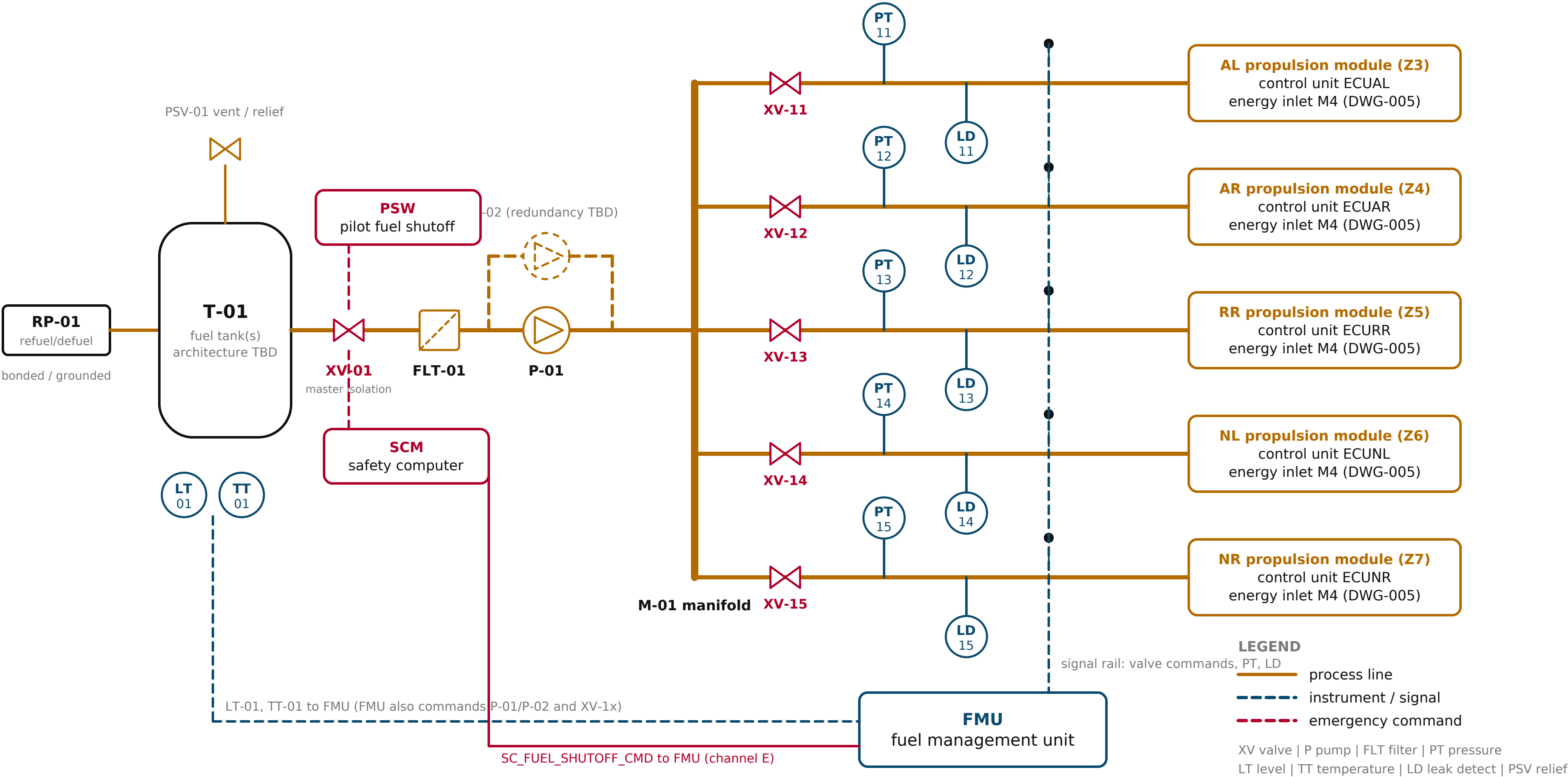
BOLTE HFP-X	Structural load paths Mounting interfaces	
HFPX-TDATA-DWG-007 Drawing ID	Sheet 7 of 13 Scale: NTS	
CONCEPT - not baselined (BL-0.0) Dimensions: TBD. Not for manufacture.	2026-09-29 Approver: TBD	

8 - Fuel system piping-and-instrumentation schematic

- NOTES
1. ARCHITECTURE ONLY (HFPX-FUEL-ARC-001). Fuel type, sizes, pressures: TBD (ISS-003).

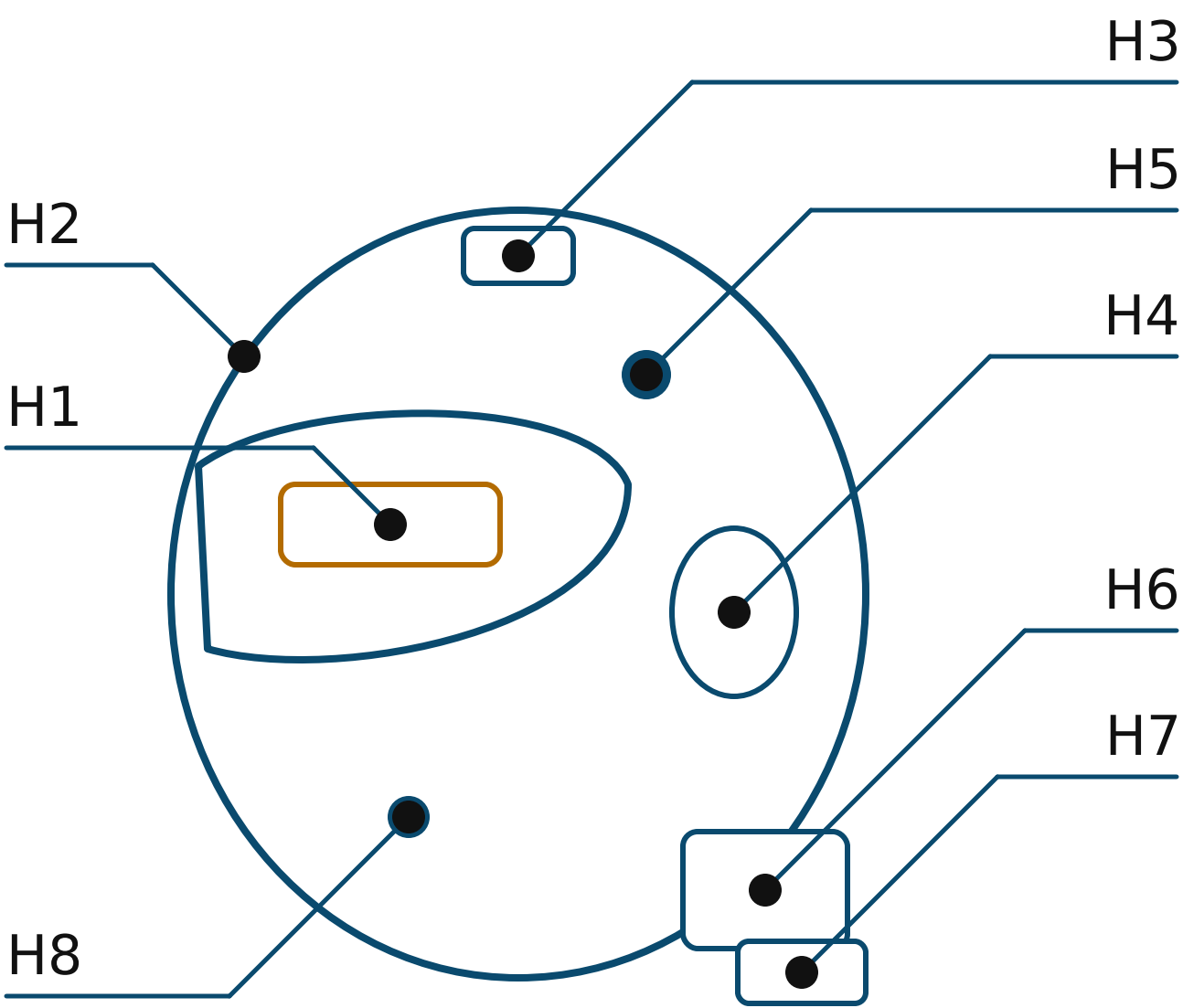
2. Valve fail position is FLIGHT-MODE DEPENDENT (ISS-009). De-energise-to-close is not assumed safe.

3. Instrument tags follow an ISA-style pattern, illustrative. Component count is A-005.



BOLTE HFP-X		Fuel system P&ID Architecture level	
HFPX-TDATA-DWG-008 Drawing ID		Sheet 8 of 13 Scale: NTS	
CONCEPT - not baselined (BL-0.0) Dimensions: TBD. Not for manufacture.		2026-09-29 Approver: TBD	

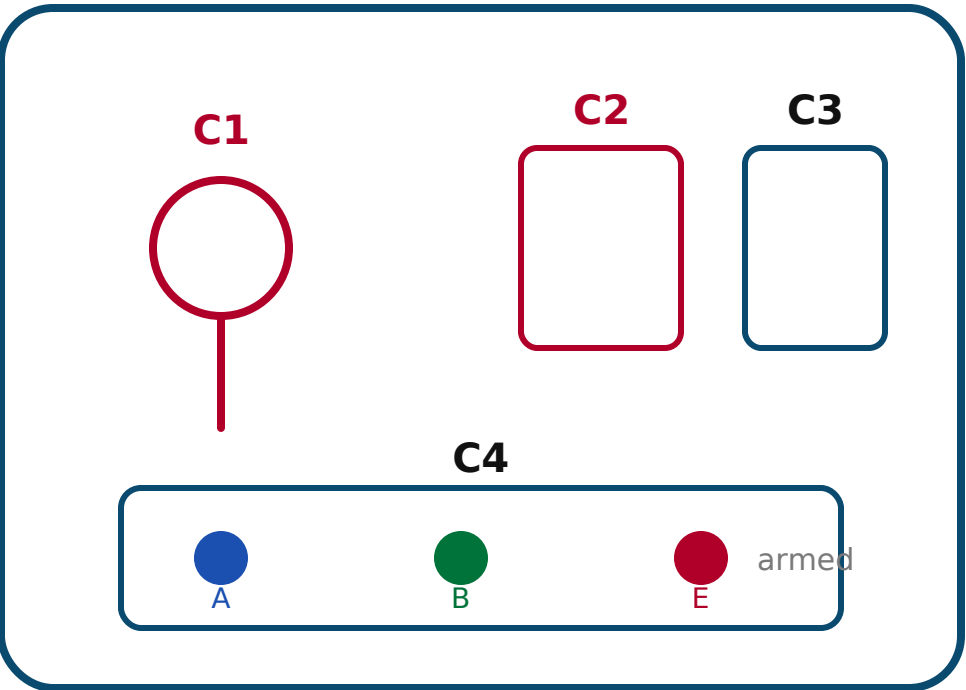
9 - Pilot interface and emergency recovery system



SMART HELMET (cutaway, left side)

- H1 HUD display / combiner
- H2 Visor and impact shell
- H3 Head-tracking sensor
- H4 Audio and microphone
- H5 Physiological sensors (pilot state, Vol 12.12)
- H6 Helmet electronics
- H7 Helmet-harness connector (power, data; breakaway TBD)
- H8 Environmental / breathing interface (TBD)

PILOT CONTROLS (chest panel concept)



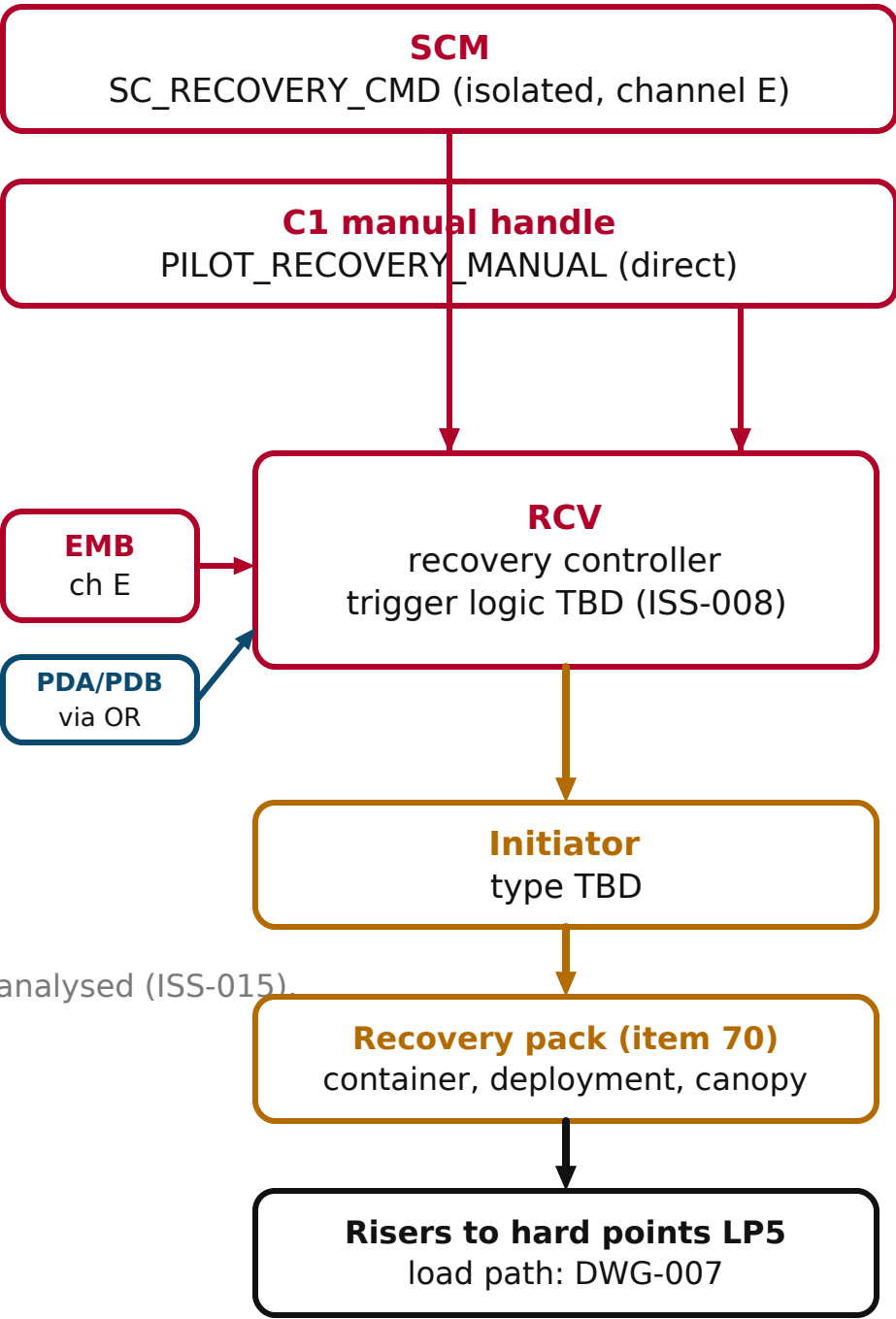
- C1 Manual recovery handle (guarded, direct to RCV)
- C2 Fuel shutoff switch (guarded, direct)
- C3 Arm / safe selector (TBD)
- C4 Status indicators: channels A, B, E; recovery armed

Reach, glove use, inadvertent activation and posture in each flight mode: unanalysed (ISS-015)

NOTES

- 1. CONCEPT drawing. No dimension, mass, thrust or angle is shown: all TBD.
- 2. Body proportions, vectors and on-body placement are ILLUSTRATIVE, not to scale.
- 3. Module set (arm L/R, rear, ankle L/R): assumption A-003. Placement: A-008.
- 4. Recovery type, trigger logic and deployment envelope: TBD (ISS-008).
- 5. Control layout and reach are unanalysed (ISS-015). Drawn for discussion only.

EMERGENCY RECOVERY SYSTEM (concept)



OPEN GAP found while drawing this sheet:
RCV has no status or health return path in wire_list.csv (only PA, PB, PE and DS connectors). It cannot report armed / healthy to the SCM or to the pilot (ISS-018).

BOLTE HFP-X		Pilot interface and recovery system	
HFPX-TDATA-DWG-009 Drawing ID		Sheet 9 of 13 Scale: NTS	
CONCEPT - not baselined (BL-0.0) Dimensions: TBD. Not for manufacture.		2026-09-29 Approver: TBD	

10 - Harness schedule (generated from wire_list.csv)

NOTES

- 1. Generated from the wiring data; edit the CSV under change control, not this drawing.
- 2. Rules DR-W-01..06 are checked by hfp_x_wiring.py check. Physical routing: TBD.

H-DAT-A-01	data, channel A	28 wires	28 connectors	15 units
Units: BB1, COM, DLG, ECUAL, ECUAR, ECUNL, ECUNR, ECURR, FCA, FCB, FMU, HLM, PIL, SCM, SNS				
H-DAT-B-01	data, channel B	28 wires	28 connectors	15 units
Units: BB2, COM, DLG, ECUAL, ECUAR, ECUNL, ECUNR, ECURR, FCA, FCB, FMU, HLM, PIL, SCM, SNS				
H-DAT-E-01	data, channel E	2 wires	2 connectors	2 units
Units: SCM, SIM				
H-DSC-E-01	discrete, channel E	18 wires	9 connectors	9 units
Units: ECUAL, ECUAR, ECUNL, ECUNR, ECURR, FMU, PIL, RCV, SCM				
H-PWR-A-01	power, channel A	30 wires	17 connectors	16 units
Units: COM, DLG, ECUAL, ECUAR, ECUNL, ECUNR, ECURR, EMB, FCA, FMU, HLM, PDA, RCV, SCM, SNS, SRCA				
H-PWR-B-01	power, channel B	24 wires	14 connectors	13 units
Units: COM, ECUAL, ECUAR, ECUNL, ECUNR, ECURR, FCB, FMU, PDB, RCV, SCM, SNS, SRCB				
H-PWR-E-01	power, channel E	8 wires	5 connectors	5 units
Units: COM, EMB, RCV, SCM, SIM				

Each harness carries exactly one class and one channel (DR-W-01, DR-W-02). A, B and E routes are intended to be physically separated (not machine-checkable yet).
Wire gauge, insulation, shielding and connector type: TBD (ISS-011, ISS-012, ISS-013).

BOLTE HFP-X	Harness schedule Generated from data	
HFPX-TDATA-DWG-010 Drawing ID	Sheet 10 of 13 Scale: NTS	
CONCEPT - not baselined (BL-0.0) Dimensions: TBD. Not for manufacture.	2026-09-29 Approver: TBD	

11 - Connector pinouts, sheet 1 of 3 (generated)

NOTES

- 1. Generated from connector_list.csv and wire_list.csv: every wire end appears once.
- 2. Pin numbers are ALLOCATED positions, not physical contact positions (TBD).
- 3. Right column of each card: the other end of the wire (connector:pin).

J-SRCA-OA

Primary source A (type TBD) | power, ch A | Z8 | 2 pins

1	PWR_SRCA_TO_PDA_POS	J-PDA-PA:1
2	PWR_SRCA_TO_PDA_RTN	J-PDA-PA:2

J-SRCB-OB

Primary source B (type TBD) | power, ch B | Z8 | 2 pins

1	PWR_SRCB_TO_PDB_POS	J-PDB-PB:1
2	PWR_SRCB_TO_PDB_RTN	J-PDB-PB:2

J-PDA-OA

Power distribution unit A | power, ch A | Z8 | 28 pins

1	PWR_PDA_TO_FCA_POS	J-FCA-PA:1
2	PWR_PDA_TO_FCA_RTN	J-FCA-PA:2
3	PWR_PDA_TO_SCM_POS	J-SCM-PA:1
4	PWR_PDA_TO_SCM_RTN	J-SCM-PA:2
5	PWR_PDA_TO_SNS_POS	J-SNS-PA:1
6	PWR_PDA_TO_SNS_RTN	J-SNS-PA:2
7	PWR_PDA_TO_FMU_POS	J-FMU-PA:1
8	PWR_PDA_TO_FMU_RTN	J-FMU-PA:2
9	PWR_PDA_TO_COM_POS	J-COM-PA:1
10	PWR_PDA_TO_COM_RTN	J-COM-PA:2
11	PWR_PDA_TO_HLM_POS	J-HLM-PA:1
12	PWR_PDA_TO_HLM_RTN	J-HLM-PA:2
13	PWR_PDA_TO_DLG_POS	J-DLG-PA:1
14	PWR_PDA_TO_DLG_RTN	J-DLG-PA:2
15	PWR_PDA_TO_RCV_POS	J-RCV-PA:1
16	PWR_PDA_TO_RCV_RTN	J-RCV-PA:2
17	PWR_PDA_TO_EMB_POS	J-EMB-PA:1
18	PWR_PDA_TO_EMB_RTN	J-EMB-PA:2
19	PWR_PDA_TO_ECUAL_POS	J-ECUAL-PA:1
20	PWR_PDA_TO_ECUAL_RTN	J-ECUAL-PA:2
21	PWR_PDA_TO_ECUAR_POS	J-ECUAR-PA:1
22	PWR_PDA_TO_ECUAR_RTN	J-ECUAR-PA:2
23	PWR_PDA_TO_ECURRE_POS	J-ECURR-PA:1
24	PWR_PDA_TO_ECURRE_RTN	J-ECURR-PA:2
25	PWR_PDA_TO_ECUNL_POS	J-ECUNL-PA:1
26	PWR_PDA_TO_ECUNL_RTN	J-ECUNL-PA:2
27	PWR_PDA_TO_ECUNR_POS	J-ECUNR-PA:1
28	PWR_PDA_TO_ECUNR_RTN	J-ECUNR-PA:2

J-PDA-PA

Power distribution unit A | power, ch A | Z8 | 2 pins

1	PWR_SRCA_TO_PDA_POS	J-SRCA-OA:1
2	PWR_SRCA_TO_PDA_RTN	J-SRCA-OA:2

J-PDB-OB

Power distribution unit B | power, ch B | Z8 | 22 pins

1	PWR_PDB_TO_FCB_POS	J-FCB-PB:1
2	PWR_PDB_TO_FCB_RTN	J-FCB-PB:2
3	PWR_PDB_TO_SCM_POS	J-SCM-PB:1
4	PWR_PDB_TO_SCM_RTN	J-SCM-PB:2
5	PWR_PDB_TO_SNS_POS	J-SNS-PB:1
6	PWR_PDB_TO_SNS_RTN	J-SNS-PB:2
7	PWR_PDB_TO_FMU_POS	J-FMU-PB:1
8	PWR_PDB_TO_FMU_RTN	J-FMU-PB:2
9	PWR_PDB_TO_COM_POS	J-COM-PB:1
10	PWR_PDB_TO_COM_RTN	J-COM-PB:2
11	PWR_PDB_TO_RCV_POS	J-RCV-PB:1
12	PWR_PDB_TO_RCV_RTN	J-RCV-PB:2
13	PWR_PDB_TO_ECUAL_POS	J-ECUAL-PB:1
14	PWR_PDB_TO_ECUAL_RTN	J-ECUAL-PB:2
15	PWR_PDB_TO_ECUAR_POS	J-ECUAR-PB:1
16	PWR_PDB_TO_ECUAR_RTN	J-ECUAR-PB:2
17	PWR_PDB_TO_ECURRE_POS	J-ECURR-PB:1
18	PWR_PDB_TO_ECURRE_RTN	J-ECURR-PB:2
19	PWR_PDB_TO_ECUNL_POS	J-ECUNL-PB:1
20	PWR_PDB_TO_ECUNL_RTN	J-ECUNL-PB:2
21	PWR_PDB_TO_ECUNR_POS	J-ECUNR-PB:1
22	PWR_PDB_TO_ECUNR_RTN	J-ECUNR-PB:2

J-PDB-PB

Power distribution unit B | power, ch B | Z8 | 2 pins

1	PWR_SRCB_TO_PDB_POS	J-SRCB-OB:1
2	PWR_SRCB_TO_PDB_RTN	J-SRCB-OB:2

J-EMB-OE

Emergency battery | power, ch E | Z8 | 8 pins

1	PWR_EMB_TO_SCM_POS	J-SCM-PE:1
2	PWR_EMB_TO_SCM_RTN	J-SCM-PE:2
3	PWR_EMB_TO_RCV_POS	J-RCV-PE:1
4	PWR_EMB_TO_RCV_RTN	J-RCV-PE:2
5	PWR_EMB_TO_COM_POS	J-COM-PE:1
6	PWR_EMB_TO_COM_RTN	J-COM-PE:2
7	PWR_EMB_TO_SIM_POS	J-SIM-PE:1
8	PWR_EMB_TO_SIM_RTN	J-SIM-PE:2

J-EMB-PA

Emergency battery | power, ch A | Z8 | 2 pins

1	PWR_PDA_TO_EMB_POS	J-PDA-OA:17
2	PWR_PDA_TO_EMB_RTN	J-PDA-OA:18

J-FCA-D1

Flight computer A | data, ch A | Z8 | 2 pins

1	DATA_BUS1_FCA_H	J-BB1-FCA:1
2	DATA_BUS1_FCA_L	J-BB1-FCA:2

J-FCA-D2

Flight computer A | data, ch B | Z8 | 2 pins

1	DATA_BUS2_FCA_H	J-BB2-FCA:1
2	DATA_BUS2_FCA_L	J-BB2-FCA:2

J-FCA-PA

Flight computer A | power, ch A | Z8 | 2 pins

1	PWR_PDA_TO_FCA_POS	J-PDA-OA:1
2	PWR_PDA_TO_FCA_RTN	J-PDA-OA:2

J-FCB-D1

Flight computer B | data, ch A | Z8 | 2 pins

1	DATA_BUS1_FCB_H	J-BB1-FCB:1
2	DATA_BUS1_FCB_L	J-BB1-FCB:2

J-FCB-D2

Flight computer B | data, ch B | Z8 | 2 pins

1	DATA_BUS2_FCB_H	J-BB2-FCB:1
2	DATA_BUS2_FCB_L	J-BB2-FCB:2

J-FCB-PB

Flight computer B | power, ch B | Z8 | 2 pins

1	PWR_PDB_TO_FCB_POS	J-PDB-OB:1
2	PWR_PDB_TO_FCB_RTN	J-PDB-OB:2

J-SCM-D1

Safety computer | data, ch A | Z8 | 2 pins

1	DATA_BUS1_SCM_H	J-BB1-SCM:1
2	DATA_BUS1_SCM_L	J-BB1-SCM:2

J-SCM-D2

Safety computer | data, ch B | Z8 | 2 pins

1	DATA_BUS2_SCM_H	J-BB2-SCM:1
2	DATA_BUS2_SCM_L	J-BB2-SCM:2

J-SCM-DE

Safety computer | data, ch E | Z8 | 2 pins

1	DATA_SIM_SCM_H	J-SIM-DE:1
2	DATA_SIM_SCM_L	J-SIM-DE:2

J-SCM-DS

Safety computer | discrete, ch E | Z8 | 14 pins

1	SC_PERMIT_ECUAL	J-ECUAL-DS:1
2	SC_PERMIT_ECUAL_RTN	J-ECUAL-DS:2
3	SC_PERMIT_ECUAR	J-ECUAR-DS:1
4	SC_PERMIT_ECUAR_RTN	J-ECUAR-DS:2
5	SC_PERMIT_ECURRE	J-ECURR-DS:1
6	SC_PERMIT_ECURRE_RTN	J-ECURR-DS:2
7	SC_PERMIT_ECUNL	J-ECUNL-DS:1
8	SC_PERMIT_ECUNL_RTN	J-ECUNL-DS:2
9	SC_PERMIT_ECUNR	J-ECUNR-DS:1
10	SC_PERMIT_ECUNR_RTN	J-ECUNR-DS:2
11	SC_FUEL_SHUTOFF_CMD	J-FMU-DS:1
12	SC_FUEL_SHUTOFF_CMD_RTN	J-FMU-DS:2
13	SC_RECOVERY_CMD	J-RCV-DS:1
14	SC_RECOVERY_CMD_RTN	J-RCV-DS:2

J-SCM-PA

Safety computer | power, ch A | Z8 | 2 pins

1	PWR_PDA_TO_SCM_POS	J-PDA-OA:3
2	PWR_PDA_TO_SCM_RTN	J-PDA-OA:4

J-SCM-PB

Safety computer | power, ch B | Z8 | 2 pins

1	PWR_PDB_TO_SCM_POS	J-PDB-OB:3
2	PWR_PDB_TO_SCM_RTN	J-PDB-OB:4

J-SCM-PE

Safety computer | power, ch E | Z8 | 2 pins

1	PWR_EMB_TO_SCM_POS	J-EMB-OE:1
2	PWR_EMB_TO_SCM_RTN	J-EMB-OE:2

J-SIM-DE

Safety-computer independent sensors | data, ch E | Z8 | 2 pins

1	DATA_SIM_SCM_H	J-SCM-DE:1
2	DATA_SIM_SCM_L	J-SCM-DE:2

J-SIM-PE

Safety-computer independent sensors | power, ch E | Z8 | 2 pins

1	PWR_EMB_TO_SIM_POS	J-EMB-OE:7
2	PWR_EMB_TO_SIM_RTN	J-EMB-OE:8

J-SNS-D1

Sensor cluster | data, ch A | Z8 | 2 pins

1	DATA_BUS1_SNS_H	J-BB1-SNS:1
2	DATA_BUS1_SNS_L	J-BB1-SNS:2

J-SNS-D2

Sensor cluster | data, ch B | Z8 | 2 pins

1	DATA_BUS2_SNS_H	J-BB2-SNS:1
2	DATA_BUS2_SNS_L	J-BB2-SNS:2

J-SNS-PA

Sensor cluster | power, ch A | Z8 | 2 pins

1	PWR_PDA_TO_SNS_POS	J-PDA-OA:5
2	PWR_PDA_TO_SNS_RTN	J-PDA-OA:6

J-SNS-PB

Sensor cluster | power, ch B | Z8 | 2 pins

1	PWR_PDB_TO_SNS_POS	J-PDB-OB:5
2	PWR_PDB_TO_SNS_RTN	J-PDB-OB:6

J-ECUAL-D1

Propulsion control unit - arm left | data, ch A | Z3 | 2 pins

1	DATA_BUS1_ECUAL_H	J-BB1-ECUAL:1
2	DATA_BUS1_ECUAL_L	J-BB1-ECUAL:2

J-ECUAL-D2

Propulsion control unit - arm left | data, ch B | Z3 | 2 pins

1	DATA_BUS2_ECUAL_H	J-BB2-ECUAL:1
2	DATA_BUS2_ECUAL_L	J-BB2-ECUAL:2

J-ECUAL-DS

Propulsion control unit - arm left | discrete, ch E | Z3 | 2 pins

1	SC_PERMIT_ECUAL	J-SCM-DS:1
2	SC_PERMIT_ECUAL_RTN	J-SCM-DS:2

BOLTE HFP-X	Connector pinouts sheet 1 of 3	
HFPX-TDATA-DWG-101 Drawing ID	Sheet 11 of 13 Scale: NTS	
CONCEPT - not baselined (BL-0.0) Dimensions: TBD. Not for manufacture.	2026-09-29 Approver: TBD	

12 - Connector pinouts, sheet 2 of 3 (generated)

NOTES

- 1. Generated from connector_list.csv and wire_list.csv: every wire end appears once.
- 2. Pin numbers are ALLOCATED positions, not physical contact positions (TBD).
- 3. Right column of each card: the other end of the wire (connector:pin).

J-ECUAL-PA Propulsion control unit - arm left power, ch A Z3 2 pins 1 PWR_PDA_TO_ECUAL_POS J-PDA-OA:19 2 PWR_PDA_TO_ECUAL_RTN J-PDA-OA:20	J-ECURR-D2 Propulsion control unit - rear data, ch B Z5 2 pins 1 DATA_BUS2_ECURRE_H J-BB2-ECURR:1 2 DATA_BUS2_ECURRE_L J-BB2-ECURR:2	J-ECUNL-PB Propulsion control unit - ankle left power, ch B Z6 2 pins 1 PWR_PDB_TO_ECUNL_POS J-PDB-OB:19 2 PWR_PDB_TO_ECUNL_RTN J-PDB-OB:20	J-FMU-DS Fuel management unit discrete, ch E Z9 4 pins 1 SC_FUEL_SHUTOFF_CMD J-SCM-DS:11 2 SC_FUEL_SHUTOFF_CMD_RTN J-SCM-DS:12 3 PILOT_FUEL_SHUTOFF J-PIL-DS:3 4 PILOT_FUEL_SHUTOFF_RTN J-PIL-DS:4	J-COM-D2 Comms / telemetry unit data, ch B Z8 2 pins 1 DATA_BUS2_COM_H J-BB2-COM:1 2 DATA_BUS2_COM_L J-BB2-COM:2
J-ECUAL-PB Propulsion control unit - arm left power, ch B Z3 2 pins 1 PWR_PDB_TO_ECUAL_POS J-PDB-OB:13 2 PWR_PDB_TO_ECUAL_RTN J-PDB-OB:14	J-ECURR-DS Propulsion control unit - rear discrete, ch E Z5 2 pins 1 SC_PERMIT_ECURRE J-SCM-DS:5 2 SC_PERMIT_ECURRE_RTN J-SCM-DS:6	J-ECUNR-D1 Propulsion control unit - ankle right data, ch A Z7 2 pins 1 DATA_BUS1_ECUNR_H J-BB1-ECUNR:1 2 DATA_BUS1_ECUNR_L J-BB1-ECUNR:2	J-FMU-PA Fuel management unit power, ch A Z9 2 pins 1 PWR_PDA_TO_FMU_POS J-PDA-OA:7 2 PWR_PDA_TO_FMU_RTN J-PDA-OA:8	J-COM-PA Comms / telemetry unit power, ch A Z8 2 pins 1 PWR_PDA_TO_COM_POS J-PDA-OA:9 2 PWR_PDA_TO_COM_RTN J-PDA-OA:10
J-ECUAR-D1 Propulsion control unit - arm right data, ch A Z4 2 pins 1 DATA_BUS1_ECUAR_H J-BB1-ECUAR:1 2 DATA_BUS1_ECUAR_L J-BB1-ECUAR:2	J-ECURR-PA Propulsion control unit - rear power, ch A Z5 2 pins 1 PWR_PDA_TO_ECURRE_POS J-PDA-OA:23 2 PWR_PDA_TO_ECURRE_RTN J-PDA-OA:24	J-ECUNR-D2 Propulsion control unit - ankle right data, ch B Z7 2 pins 1 DATA_BUS2_ECUNR_H J-BB2-ECUNR:1 2 DATA_BUS2_ECUNR_L J-BB2-ECUNR:2	J-FMU-PB Fuel management unit power, ch B Z9 2 pins 1 PWR_PDB_TO_FMU_POS J-PDB-OB:7 2 PWR_PDB_TO_FMU_RTN J-PDB-OB:8	J-COM-PB Comms / telemetry unit power, ch B Z8 2 pins 1 PWR_PDB_TO_COM_POS J-PDB-OB:9 2 PWR_PDB_TO_COM_RTN J-PDB-OB:10
J-ECUAR-D2 Propulsion control unit - arm right data, ch B Z4 2 pins 1 DATA_BUS2_ECUAR_H J-BB2-ECUAR:1 2 DATA_BUS2_ECUAR_L J-BB2-ECUAR:2	J-ECURR-PB Propulsion control unit - rear power, ch B Z5 2 pins 1 PWR_PDB_TO_ECURRE_POS J-PDB-OB:17 2 PWR_PDB_TO_ECURRE_RTN J-PDB-OB:18	J-ECUNR-DS Propulsion control unit - ankle right discrete, ch E Z7 2 pins 1 SC_PERMIT_ECUNR J-SCM-DS:9 2 SC_PERMIT_ECUNR_RTN J-SCM-DS:10	J-RCV-DS Recovery controller discrete, ch E Z8 4 pins 1 SC_RECOVERY_CMD J-SCM-DS:13 2 SC_RECOVERY_CMD_RTN J-SCM-DS:14 3 PILOT_RECOVERY_MANUAL J-PIL-DS:1 4 PILOT_RECOVERY_MANUAL_RTN J-PIL-DS:2	J-COM-PE Comms / telemetry unit power, ch E Z8 2 pins 1 PWR_EMB_TO_COM_POS J-EMB-OE:5 2 PWR_EMB_TO_COM_RTN J-EMB-OE:6
J-ECUAR-DS Propulsion control unit - arm right discrete, ch E Z4 2 pins 1 SC_PERMIT_ECUAR J-SCM-DS:3 2 SC_PERMIT_ECUAR_RTN J-SCM-DS:4	J-ECUNL-D1 Propulsion control unit - ankle left data, ch A Z6 2 pins 1 DATA_BUS1_ECUNL_H J-BB1-ECUNL:1 2 DATA_BUS1_ECUNL_L J-BB1-ECUNL:2	J-ECUNR-PA Propulsion control unit - ankle right power, ch A Z7 2 pins 1 PWR_PDA_TO_ECUNR_POS J-PDA-OA:27 2 PWR_PDA_TO_ECUNR_RTN J-PDA-OA:28	J-RCV-PA Recovery controller power, ch A Z8 2 pins 1 PWR_PDA_TO_RCV_POS J-PDA-OA:15 2 PWR_PDA_TO_RCV_RTN J-PDA-OA:16	J-HLM-D1 Smart helmet / HUD data, ch A Z1 2 pins 1 DATA_BUS1_HLM_H J-BB1-HLM:1 2 DATA_BUS1_HLM_L J-BB1-HLM:2
J-ECUAR-PA Propulsion control unit - arm right power, ch A Z4 2 pins 1 PWR_PDA_TO_ECUAR_POS J-PDA-OA:21 2 PWR_PDA_TO_ECUAR_RTN J-PDA-OA:22	J-ECUNL-D2 Propulsion control unit - ankle left data, ch B Z6 2 pins 1 DATA_BUS2_ECUNL_H J-BB2-ECUNL:1 2 DATA_BUS2_ECUNL_L J-BB2-ECUNL:2	J-ECUNR-PB Propulsion control unit - ankle right power, ch B Z7 2 pins 1 PWR_PDB_TO_ECUNR_POS J-PDB-OB:21 2 PWR_PDB_TO_ECUNR_RTN J-PDB-OB:22	J-RCV-PB Recovery controller power, ch B Z8 2 pins 1 PWR_PDB_TO_RCV_POS J-PDB-OB:11 2 PWR_PDB_TO_RCV_RTN J-PDB-OB:12	J-HLM-D2 Smart helmet / HUD data, ch B Z1 2 pins 1 DATA_BUS2_HLM_H J-BB2-HLM:1 2 DATA_BUS2_HLM_L J-BB2-HLM:2
J-ECUAR-PB Propulsion control unit - arm right power, ch B Z4 2 pins 1 PWR_PDB_TO_ECUAR_POS J-PDB-OB:15 2 PWR_PDB_TO_ECUAR_RTN J-PDB-OB:16	J-ECUNL-DS Propulsion control unit - ankle left discrete, ch E Z6 2 pins 1 SC_PERMIT_ECUNL J-SCM-DS:7 2 SC_PERMIT_ECUNL_RTN J-SCM-DS:8	J-FMU-D1 Fuel management unit data, ch A Z9 2 pins 1 DATA_BUS1_FMU_H J-BB1-FMU:1 2 DATA_BUS1_FMU_L J-BB1-FMU:2	J-RCV-PE Recovery controller power, ch E Z8 2 pins 1 PWR_EMB_TO_RCV_POS J-EMB-OE:3 2 PWR_EMB_TO_RCV_RTN J-EMB-OE:4	J-HLM-PA Smart helmet / HUD power, ch A Z1 2 pins 1 PWR_PDA_TO_HLM_POS J-PDA-OA:11 2 PWR_PDA_TO_HLM_RTN J-PDA-OA:12
J-ECURR-D1 Propulsion control unit - rear data, ch A Z5 2 pins 1 DATA_BUS1_ECURRE_H J-BB1-ECURR:1 2 DATA_BUS1_ECURRE_L J-BB1-ECURR:2	J-ECUNL-PA Propulsion control unit - ankle left power, ch A Z6 2 pins 1 PWR_PDA_TO_ECUNL_POS J-PDA-OA:25 2 PWR_PDA_TO_ECUNL_RTN J-PDA-OA:26	J-FMU-D2 Fuel management unit data, ch B Z9 2 pins 1 DATA_BUS2_FMU_H J-BB2-FMU:1 2 DATA_BUS2_FMU_L J-BB2-FMU:2	J-COM-D1 Comms / telemetry unit data, ch A Z8 2 pins 1 DATA_BUS1_COM_H J-BB1-COM:1 2 DATA_BUS1_COM_L J-BB1-COM:2	J-DLG-D1 Data logger data, ch A Z8 2 pins 1 DATA_BUS1_DLG_H J-BB1-DLG:1 2 DATA_BUS1_DLG_L J-BB1-DLG:2

BOLTE HFP-X	Connector pinouts sheet 2 of 3	
HFPX-TDATA-DWG-102 Drawing ID	Sheet 12 of 13 Scale: NTS	
CONCEPT - not baselined (BL-0.0) Dimensions: TBD. Not for manufacture.	2026-09-29 Approver: TBD	

13 - Connector pinouts, sheet 3 of 3 (generated)

NOTES

- 1. Generated from connector_list.csv and wire_list.csv: every wire end appears once.
- 2. Pin numbers are ALLOCATED positions, not physical contact positions (TBD).
- 3. Right column of each card: the other end of the wire (connector:pin).

J-DLG-D2 Data logger data, ch B Z8 2 pins 1 DATA_BUS2_DLG_H J-BB2-DLG:1 2 DATA_BUS2_DLG_L J-BB2-DLG:2	J-BB1-ECUAR Data bus 1 backbone / junction data, ch A Z8 2 pins 1 DATA_BUS1_ECUAR_H J-ECUAR-D1:1 2 DATA_BUS1_ECUAR_L J-ECUAR-D1:2	J-BB1-PIL Data bus 1 backbone / junction data, ch A Z8 2 pins 1 DATA_BUS1_PIL_H J-PIL-D1:1 2 DATA_BUS1_PIL_L J-PIL-D1:2	J-BB2-ECUNR Data bus 2 backbone / junction data, ch B Z8 2 pins 1 DATA_BUS2_ECUNR_H J-ECUNR-D2:1 2 DATA_BUS2_ECUNR_L J-ECUNR-D2:2	J-BB2-SNS Data bus 2 backbone / junction data, ch B Z8 2 pins 1 DATA_BUS2_SNS_H J-SNS-D2:1 2 DATA_BUS2_SNS_L J-SNS-D2:2
J-DLG-PA Data logger power, ch A Z8 2 pins 1 PWR_PDA_TO_DLG_POS J-PDA-OA:13 2 PWR_PDA_TO_DLG_RTN J-PDA-OA:14	J-BB1-ECUNL Data bus 1 backbone / junction data, ch A Z8 2 pins 1 DATA_BUS1_ECUNL_H J-ECUNL-D1:1 2 DATA_BUS1_ECUNL_L J-ECUNL-D1:2	J-BB1-SCM Data bus 1 backbone / junction data, ch A Z8 2 pins 1 DATA_BUS1_SCM_H J-SCM-D1:1 2 DATA_BUS1_SCM_L J-SCM-D1:2	J-BB2-ECURR Data bus 2 backbone / junction data, ch B Z8 2 pins 1 DATA_BUS2_ECURR_H J-ECURR-D2:1 2 DATA_BUS2_ECURR_L J-ECURR-D2:2	
J-PIL-D1 Pilot inputs and manual handles data, ch A Z2 2 pins 1 DATA_BUS1_PIL_H J-BB1-PIL:1 2 DATA_BUS1_PIL_L J-BB1-PIL:2	J-BB1-ECUNR Data bus 1 backbone / junction data, ch A Z8 2 pins 1 DATA_BUS1_ECUNR_H J-ECUNR-D1:1 2 DATA_BUS1_ECUNR_L J-ECUNR-D1:2	J-BB1-SNS Data bus 1 backbone / junction data, ch A Z8 2 pins 1 DATA_BUS1_SNS_H J-SNS-D1:1 2 DATA_BUS1_SNS_L J-SNS-D1:2	J-BB2-FCA Data bus 2 backbone / junction data, ch B Z8 2 pins 1 DATA_BUS2_FCA_H J-FCA-D2:1 2 DATA_BUS2_FCA_L J-FCA-D2:2	
J-PIL-D2 Pilot inputs and manual handles data, ch B Z2 2 pins 1 DATA_BUS2_PIL_H J-BB2-PIL:1 2 DATA_BUS2_PIL_L J-BB2-PIL:2	J-BB1-ECURR Data bus 1 backbone / junction data, ch A Z8 2 pins 1 DATA_BUS1_ECURR_H J-ECURR-D1:1 2 DATA_BUS1_ECURR_L J-ECURR-D1:2	J-BB2-COM Data bus 2 backbone / junction data, ch B Z8 2 pins 1 DATA_BUS2_COM_H J-COM-D2:1 2 DATA_BUS2_COM_L J-COM-D2:2	J-BB2-FCB Data bus 2 backbone / junction data, ch B Z8 2 pins 1 DATA_BUS2_FCB_H J-FCB-D2:1 2 DATA_BUS2_FCB_L J-FCB-D2:2	
J-PIL-DS Pilot inputs and manual handles discrete, ch E Z2 4 pins 1 PILOT_RECOVERY_MANUAL J-RCV-DS:3 2 PILOT_RECOVERY_MANUAL_RTN J-RCV-DS:4 3 PILOT_FUEL_SHUTOFF J-FMU-DS:3 4 PILOT_FUEL_SHUTOFF_RTN J-FMU-DS:4	J-BB1-FCA Data bus 1 backbone / junction data, ch A Z8 2 pins 1 DATA_BUS1_FCA_H J-FCA-D1:1 2 DATA_BUS1_FCA_L J-FCA-D1:2	J-BB2-DLG Data bus 2 backbone / junction data, ch B Z8 2 pins 1 DATA_BUS2_DLG_H J-DLG-D2:1 2 DATA_BUS2_DLG_L J-DLG-D2:2	J-BB2-FMU Data bus 2 backbone / junction data, ch B Z8 2 pins 1 DATA_BUS2_FMU_H J-FMU-D2:1 2 DATA_BUS2_FMU_L J-FMU-D2:2	
J-BB1-COM Data bus 1 backbone / junction data, ch A Z8 2 pins 1 DATA_BUS1_COM_H J-COM-D1:1 2 DATA_BUS1_COM_L J-COM-D1:2	J-BB1-FCB Data bus 1 backbone / junction data, ch A Z8 2 pins 1 DATA_BUS1_FCB_H J-FCB-D1:1 2 DATA_BUS1_FCB_L J-FCB-D1:2	J-BB2-ECUAL Data bus 2 backbone / junction data, ch B Z8 2 pins 1 DATA_BUS2_ECUAL_H J-ECUAL-D2:1 2 DATA_BUS2_ECUAL_L J-ECUAL-D2:2	J-BB2-HLM Data bus 2 backbone / junction data, ch B Z8 2 pins 1 DATA_BUS2_HLM_H J-HLM-D2:1 2 DATA_BUS2_HLM_L J-HLM-D2:2	
J-BB1-DLG Data bus 1 backbone / junction data, ch A Z8 2 pins 1 DATA_BUS1_DLG_H J-DLG-D1:1 2 DATA_BUS1_DLG_L J-DLG-D1:2	J-BB1-FMU Data bus 1 backbone / junction data, ch A Z8 2 pins 1 DATA_BUS1_FMU_H J-FMU-D1:1 2 DATA_BUS1_FMU_L J-FMU-D1:2	J-BB2-ECUAR Data bus 2 backbone / junction data, ch B Z8 2 pins 1 DATA_BUS2_ECUAR_H J-ECUAR-D2:1 2 DATA_BUS2_ECUAR_L J-ECUAR-D2:2	J-BB2-PIL Data bus 2 backbone / junction data, ch B Z8 2 pins 1 DATA_BUS2_PIL_H J-PIL-D2:1 2 DATA_BUS2_PIL_L J-PIL-D2:2	
J-BB1-ECUAL Data bus 1 backbone / junction data, ch A Z8 2 pins 1 DATA_BUS1_ECUAL_H J-ECUAL-D1:1 2 DATA_BUS1_ECUAL_L J-ECUAL-D1:2	J-BB1-HLM Data bus 1 backbone / junction data, ch A Z8 2 pins 1 DATA_BUS1_HLM_H J-HLM-D1:1 2 DATA_BUS1_HLM_L J-HLM-D1:2	J-BB2-ECUNL Data bus 2 backbone / junction data, ch B Z8 2 pins 1 DATA_BUS2_ECUNL_H J-ECUNL-D2:1 2 DATA_BUS2_ECUNL_L J-ECUNL-D2:2	J-BB2-SCM Data bus 2 backbone / junction data, ch B Z8 2 pins 1 DATA_BUS2_SCM_H J-SCM-D2:1 2 DATA_BUS2_SCM_L J-SCM-D2:2	

BOLTE HFP-X	Connector pinouts sheet 3 of 3	
HFPX-TDATA-DWG-103 Drawing ID	Sheet 13 of 13 Scale: NTS	
CONCEPT - not baselined (BL-0.0) Dimensions: TBD. Not for manufacture.	2026-09-29 Approver: TBD	